

Systematic Errors in DC Offset Measurement and Mitigation at Neurostimulation Electrodes: Causes, Implications and Solutions

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Abstract

Objective. Neuromodulation therapies are becoming increasingly common as a therapeutic strategy to treat neurological disorders and deficits. Despite their widespread use, the frameworks used to evaluate device longevity and electrochemical safety are limited, producing variable results for platinum (Pt) electrodes across studies. In addition, next-generation neuromodulation electrodes like thin-film microelectrode arrays (TFMEAs) are more prone to failure by virtue of their smaller size, higher edge to bulk ratios, and greater sensitivity to electrochemical dissolution processes. Two potential contributors to the variability in reported electrochemical safety limits and device performance are particularly sensitive to scaling electrodes to smaller geometries, namely: 1) insufficient measurement device input impedance and 2) unmitigated stimulator leakage currents.

Approach. To explore these effects, we first characterize the impact of electrode size (1.27×10^{-4} to 7.85×10^{-3} cm²) on apparent open circuit potential (V_{aOCP}) and interpulse potential (V_{IPP}) measurements at Pt electrodes across measurement instrumentation configurations. Next, we evaluated the effectiveness of commonly used DC mitigation strategies, including stimulation with capacitive coupling (CC), and with capacitive coupling combined with cathode/anode shorting after each pulse (CC+ES), as a function of electrode size. **Main Results.** Without proper DC mitigation, insufficient measurement input impedance leads to an underestimation of DC offsets (ΔV_{IPP}) and, consequently, overestimation of charge injection limit (Q_{inj}) at Pt electrodes. In contrast, when DC offsets are properly mitigated and the input impedance of the measurement device is sufficient, the experimentally derived charge injection limits for Pt electrodes are larger than what has been reported historically. **Significance.** By identifying the instrumentation-related confounds biasing measurements of electrochemical safety limits, this work provides a more consistent basis for the assessment of Q_{inj} for safety, as well as a framework for identifying and mitigating DC offset as a potential source of variability in electrophysiological findings across studies using TFMEAs.

Keywords: Voltage Transient, Charge Injection Limit, Charge Injection Capacity, DC Offsets, Potential Ratcheting, Residual Voltage, Electrochemical Safety

1. Introduction

Implantable devices to stimulate the nervous system, also known as neuromodulation, bioelectronic medicines or electroceuticals, are a staple of basic neuroscience studies and an increasingly common clinical therapy. The neuromodulation industry, including diverse therapies like cochlear implants for hearing restoration and deep brain stimulation (DBS) for Parkinson's disease, is projected to exceed \$10 billion in yearly revenue by 2030 [1]. With the implanted patient population increasing rapidly, designing neuromodulation devices that maximize therapeutic efficacy while ensuring device longevity remains a critical challenge.

For decades, platinum (Pt) and platinum alloys have served as the dominant electrode materials for neural stimulation devices due to their chemical inertness and corrosion tolerance. Nonetheless, these materials are not immune to degradation. Within the cochlear implant literature, a number of post-mortem analyses have revealed Pt deposits in adjacent tissue, indicating that Pt macroelectrodes undergo measurable corrosion under clinically relevant charge densities [2,3]. Historically, these findings have been dismissed as clinically insignificant because they have not been linked to adverse health outcomes or persistent device failures [4]. Nevertheless, as devices become smaller, the amount of dissolution that is tolerable before catastrophic device failure decreases drastically.

Efforts to reduce surgical invasiveness and improve the spatial precision of neural activation have led to the development and use of thin-film microelectrode arrays (TFMEAs) in both preclinical and clinical neuromodulation devices. These advancements have highlighted a unique limitation of TFMEAs: their susceptibility to corrosion-related failures compared to traditional macroelectrodes [5,6]. This vulnerability arises from multiple effects inherent to miniaturization. For a given applied current, smaller electrodes produce much higher charge densities, which increase the likelihood of undesirable electrochemical reactions associated with water electrolysis and corrosion. In addition, TFMEAs have limited material volume and higher edge-to-bulk ratios, making them more vulnerable to premature failure than macroelectrodes [5–7]. Despite numerous reports of premature failure of TFMEAs, the precise mechanisms underlying these failures remain poorly understood. Additionally, the safety frameworks used to define safe stimulation parameters were developed using Pt macroelectrodes and do not account well for the increased electrochemical vulnerability of TFMEAs [8–11].

The current frameworks used to evaluate the electrochemical and biological safety of neuromodulation devices were either derived from in-vitro voltage transients (VTs) [12] or phenomenological observations from in-vivo animal studies [10,13]. These frameworks have resulted in widely variable safety limits across studies ranging from 20-300 $\mu\text{C}/\text{cm}^2$ for Pt macroelectrodes [12,14,15]. To adhere to these limits in practice, most clinical neuromodulation devices utilize charge-balanced, current-controlled waveforms to elicit neural activity while minimizing the risk of electrode corrosion and the accumulation of toxic byproducts [16]. However, the application of charge-balanced current waveforms can induce measurable shifts in the interpulse potential (V_{IPP}) of the stimulation electrode, a phenomenon referred to as potential ratcheting, residual voltage, or the accumulation of a direct current (DC) offset [17–20]. The presence of significant DC offsets is known to accelerate electrode corrosion, alter local pH, and cause tissue damage [18,21–23]. Moreover, significant DC offsets may indicate the presence of unmitigated stimulator leakage currents, which themselves may activate or inhibit nervous tissue, resulting in altered therapeutic efficacy [24]. Therefore, ensuring device safety, efficacy and longevity requires both precise quantification and effective mitigation of DC offset accumulation.

Although the consequences of persistent DC offsets have been extensively studied, the influence of instrumentation on experimentally derived charge injection limits (Q_{inj}) in the presence of these offsets has not been properly investigated. Many studies reporting Q_{inj} from VT analysis either use measurement devices with insufficient input impedance (i.e. oscilloscopes) or fail to adequately minimize stimulator leakage currents. **We hypothesize that these instrumentation-related confounds have systematically misrepresented**

experimentally derived Q_{inj} thereby contributing to the wide variability in reported electrochemical safety limits across the neuromodulation literature.

To evaluate the extent of these contributions, we first characterize the impact of electrode size (ranging from 1.27×10^{-4} to $7.85 \times 10^{-3} \text{ cm}^2$) on apparent open circuit potential (V_{aOCP}) measurements at platinum electrodes across measurement instrumentation configurations. We demonstrate that the mismeasurement of the DC offsets due to probe-loading can dramatically influence experimentally determined limits for safe stimulation as assessed by voltage transient (VT) analysis, especially with progressively smaller electrodes. We then evaluate commonly used DC mitigation strategies, including stimulation with capacitive coupling, and with capacitive coupling combined with cathode/anode shorting after each pulse, as a function of electrode size. Collectively, these results show how DC offsets that develop on small electrodes are easily mismeasured when an oscilloscope is used. When appropriately measured, common DC mitigation strategies, like the use of a DC blocking capacitor, are often insufficient in isolation to completely mitigate stimulator leakage currents at small electrodes. Together, these data help explain the wide variation in reported safety limits, as well as the inconsistency in evoked electrophysiological responses observed across the literature [6,25–28]. Based on these results, we outline critical data that should be reported in every study, whether it be intended to assess safety for human use, or to improve rigor and reproducibility across fundamental neuroscience studies.

2. Methods

2.1 Electrode Fabrication and Polishing

90-10 platinum-iridium (Pt-Ir) disk electrodes of three different diameters; 1 mm, 0.250 mm and 0.127 mm (Surepure Chemicals PN:1884, Alfa Aesar PN:39383, AM Systems PN: 778000, respectively) with surface areas of 7.85×10^{-3} , 4.91×10^{-4} and $1.27 \times 10^{-4} \text{ cm}^2$, were prepared. For each electrode, a 76.2 mm copper wire was soldered to 5 mm length of uncoated platinum wire. Soldered connections were cleaned via ultrasonic cleaner (Fisherbrand, PN:FB11201) at a frequency of 37 Hz and a power of 100% in DI water with a few drops of surfactant for 10 minutes. Post cleaning, the platinum electrode protrusion was encased in EPO TEK 301 (PN:FB194-ND) epoxy and allowed to cure. Once cured, electrodes were polished to 10 μm grit and cleaned once more ultrasonically using the same parameters. All electrodes were also polished and cleaned periodically between voltage transient experiments to standardize the surface condition of the electrodes [29].

2.2 Electrochemical Characterization

Electrochemical impedance spectroscopy (EIS) was performed on all samples. Additionally, a set of apparent open circuit potential (V_{aOCP}) and voltage transient (VT) measurements were collected for each of the platinum electrodes ($n = 12$) at 4 different effective input impedances: 1 $\text{M}\Omega$, 10 $\text{M}\Omega$, 10 $\text{G}\Omega$, and 10 $\text{T}\Omega$. All electrochemical measurements were performed in 1X (0.01 M) phosphate buffered saline (P38135, Sigma-Aldrich) utilizing a 3-electrode setup against a large platinum counter electrode (Metrohm, 1 cm^2) and a single junction Ag|AgCl reference electrode (BASi, PN: MF-2056) filled with 3M KCl. All electrochemical measurements were performed at room temperature. (**Figure 1(a)**)

2.2.1 Electrochemical Impedance Spectroscopy (EIS) Measurements

EIS measurements were performed using an Autolab potentiostat (PG-STAT12, Metrohm) with a built-in frequency response analyzer (FRA2), EIS measurements were performed by applying 10 mV sine waves, at 0V and the open circuit potential (OCP) relative to a Ag|AgCl reference electrode. Impedance spectra were collected from 0.1 Hz to 100 kHz, spaced at 10 points per decade and subsequently fitted with an equivalent Randles circuit

to determine charge transfer resistance (R_{ct}), solution resistance (R_s) and constant phase element (CPE) values. (Figure 1(b), Supplementary Figure 1, Supplementary Table 1, Supplementary Table 2)

2.2.2 Apparent Open Circuit Potential Measurements at Equilibrium

For the V_{aOCP} measurements, electrodes of 1 mm ($n=4$), 0.25 mm ($n=4$) and 0.127 mm ($n=4$) diameter were placed in 1X PBS and connected to a Keithley DAQ6510 multimeter (Figure 1(c)). The three electrode sizes tested were selected based off pilot studies and were intended to span the range where we begin to observe appreciable probe loading effects using common oscilloscopes (i.e. typically with input impedances of 1-10 M Ω). The DAQ6510 multimeter was selected because it has a maximum input impedance greater than 10 G Ω as well as the ability to conduct measurements at 10 M Ω input impedance. 1 M Ω V_{aOCP} measurements were obtained by adding a 1M Ω oscilloscope probe in parallel with the DAQ6510. 10 T Ω V_{aOCP} measurements were obtained by using the guard port of a Keithley 6221 current source. The V_{aOCP} of each electrode was measured with respect to a Ag|AgCl electrode and allowed to stabilize until a 0.5 mV/s change in potential was observed. Three successive measurements were performed on each electrode and then averaged to produce the value presented in Figure 1(d).

2.2.3 Voltage Transient Measurements

VT measurements were performed at all input impedances by applying cathodic-leading, symmetric, charge-balanced biphasic waveforms (200 μ s/phase, 50 μ s interphase, 30 Hz, 750 pulses) using a Keithley 6221 current source. This current source was selected for its particularly high output impedance (>10 T Ω) which is essential to isolate the parasitics introduced strictly from the measurement instrumentation. Twenty logarithmically spaced charge densities between 2 and 600 μ C/cm² were applied to each electrode and the resulting voltage response was measured (Supplementary Table 3). These parameters were inspired by the seminal paper by Robblee, et al., presented in 1990 [12]. Electrodes were allowed to stabilize between trials until a 0.5 mV/s change in V_{IPP} was observed. The compliance voltage of the Keithley 6221 was set to 20V for all experiments. The interfacial electrode polarization (i.e. E_{mc} and E_{ma}) were calculated on the last pulse of each charge density applied by subtracting the access voltage (V_a) to isolate the polarization of the electrode-electrolyte interface with respect to Ag|AgCl [30,31]. The charge injection limit (Q_{inj}) in this study is defined as the charge density in which either E_{ma} or E_{mc} exceeded 0.8 and -0.6 V, respectively. (Figure 3(a)) Alternatively, in some studies, this value has also been referred to as the charge injection capacity [12,32]. It is recommended to perform a validation of the V_a subtraction to evaluate the performance of the Q_{inj} determination methodology using a secondary technique [33]. As validation, the calculated access voltages were converted into an equivalent value of R_s using the applied current. These values of R_s were then compared to the R_s values obtained from the EIS Randles fit (Supplementary Table 4, Supplementary Figure 3).

2.2.4 Implementation of Common DC Mitigation Strategies

VT measurements were also collected at the input impedances of 1 M Ω , 10 M Ω , and 10 G Ω using our custom-built DC mitigation strategy which deploys two 1 μ F, low leakage current DC blocking capacitors (PN: 105PHC400K) in series with the stimulator and a high off-impedance switch (PN: DG417L) behind the capacitors and in parallel with the stimulator output. The parasitics of the DC mitigation circuitry were considered and differential trace impedances were characterized to ensure they did not impact experimental results (see Supplementary Methods). Voltage transient experiments were run using capacitive coupling (CC) only as well as capacitive coupling with electrode shorting (CC+ES) (Supplementary Figure 2). In the capacitive coupling

with electrode shorting case, the working electrode and counter electrode were shorted together following the application of each current pulse to prevent significant DC accumulation at the electrode interface. Under the capacitive coupling only condition, pulse trains were delivered to the working electrode through the DC blocking capacitors, however the working and counter electrodes were shorted between trials rather than between pulses.

2.3 Statistical Analyses

Statistical analyses were performed using Python (v3.10.11) with the SciPy library (v1.10.1). Normality was assessed using the Shapiro-Wilk test for both V_{aOCP} and Q_{inj} datasets. Results indicated a mix of normally and non-normally distributed groups; therefore, non-parametric statistical tests were used throughout. All electrochemical data are reported throughout the text as mean $\pm$ standard error of the mean (SEM).

Friedman tests were used to assess differences in V_{IPP} and Q_{inj} across input impedances. Post hoc Wilcoxon signed-rank tests compared each group to the 10 T Ω (V_{aOCP}) or 10 G Ω (Q_{inj}) condition. An additional Wilcoxon signed-rank test compared Q_{inj} with and without CC+ES. Holm-Bonferroni correction was applied to all Wilcoxon tests.

3. Results

The results presented in this manuscript highlight how measurement device input impedance, electrode size and DC mitigation strategies influence the accurate measurement and accumulation of DC offsets (ΔV_{IPP}). **Section 3.1** examines how electrode size and input impedance impacts the measurement of the V_{aOCP} . Then, in **Section 3.2**, these findings are contextualized using the voltage divider equation. In **Section 3.3**, we present voltage transient measurements and corresponding Q_{inj} collected at the same input impedances tested in **Section 3.1**. After establishing the input impedance required for reducing/eliminating the mismeasurement of ΔV_{IPP} , we use this set-up to evaluate the effectiveness of common DC mitigation strategies. The impact of input impedance and DC mitigation strategy on the calculated Q_{inj} is summarized in **Section 3.4**.

3.1 DC Electrode Impedance & Open Circuit Potential Characterization

In practice the input impedance of a measurement device should be much larger (typically by a factor of at least 100) than the impedance of the circuit being measured [34]. This minimizes probe-loading effects, preserving both the accuracy of the measurement and the underlying circuit behavior. At DC, the impedance of an electrode as modeled by a Randles circuit, is purely resistive and simplifies to $R_{dc} = R_{ct} + R_s$, where R_{ct} is the charge transfer resistance and R_s is the solution resistance. As a result, R_{dc} is usually the highest impedance of an electrode, as shown on the representative Bode plot in **Figure 1(b)** [26–28]. Therefore, the electrode DC impedance is the most relevant measure to assess appropriateness of the input impedance of the measurement device. Initially, to estimate R_{dc} of our electrodes, the value of the charge transfer and solution resistances were determined from the Randles circuit fit of the EIS spectra collected around 0V vs Ag|AgCl (**Figure 1(b)**, **Table 1**; $R_{ct} = 2.41 \times 10^5 \pm 1.93 \times 10^3$, $6.98 \times 10^6 \pm 9.35 \times 10^5$, $8.70 \times 10^6 \pm 7.30 \times 10^5 \Omega$, $R_s = 3.26 \times 10^2 \pm 5.10$, $1.46 \times 10^3 \pm 3.25 \times 10^1$, $2.83 \times 10^3 \pm 1.34 \times 10^1 M\Omega$ for 1 mm, 0.250 mm and 0.127 mm diameter electrodes, respectively). Complete Randles circuit fit parameters for each electrode are presented in **Supplementary Table 1**.

Table 1: Average Randles circuit parameters extracted from EIS around 0V vs Ag|AgCl and OCP across electrode sizes. Comparison of 0V and OCP EIS spectra across sizes is presented in **Supplementary Figure 4**.

Electrode Diameter (mm)	R _{ct} (Ω)		R _s (Ω)		Q (Ω ⁻¹ sec ^a)		n	
	0V	OCP	0V	OCP	0V	OCP	0V	OCP
1 mm	2.41E+05	2.77E+06	3.26E+02	3.26E+02	2.89E-06	2.52E-06	8.20E-01	8.35E-01
0.250 mm	6.98E+06	6.01E+07	1.46E+03	1.48E+03	1.08E-07	1.15E-07	7.71E-01	7.81E-01
0.127 mm	8.69E+06	9.79E+07	2.83E+03	2.93E+03	5.81E-08	7.29E-08	8.04E-01	8.33E-01

Reported values of open circuit potential (OCP) for platinum electrodes in 1X PBS, at pH 7.4, typically fall between 0.25V and 0.30V versus a saturated Ag|AgCl reference electrode [35–37]. **Figure 1(d)** displays the apparent steady-state OCP (V_{aOCP}) obtained by measurement devices with input impedances of 1 MΩ, 10 MΩ, 10 GΩ or 10 TΩ, respectively. At the 1 MΩ input impedance, the V_{aOCP} measurement dramatically underestimates the OCP at all electrode sizes compared to what would be expected from literature, with the V_{aOCP} at the two smallest electrode sizes approaching zero (60.4 ± 8.65 , 3.87 ± 0.715 , and 4.37 ± 0.985 mV for 1 mm, 0.250 mm and 0.127 mm diameters, respectively). At 10 MΩ, the V_{aOCP} at all electrode sizes are notably less than would be expected from literature (159 ± 11.1 , 36.6 ± 4.90 , and 32.7 ± 4.82 mV for 1 mm, 0.250 mm and 0.127 mm diameters, respectively); however, as expected, the largest electrode size with the lowest R_{dc} is much closer to established literature values. Only by using the highest input impedance configurations (10 GΩ & 10 TΩ) do the measured V_{aOCP} values, across all electrode sizes, approach the expected range from literature. When compared to the V_{aOCP} values collected at 10 TΩ, statistically significant pairwise differences were observed at the 1 MΩ, 10 MΩ, 10 GΩ input impedances across sizes.

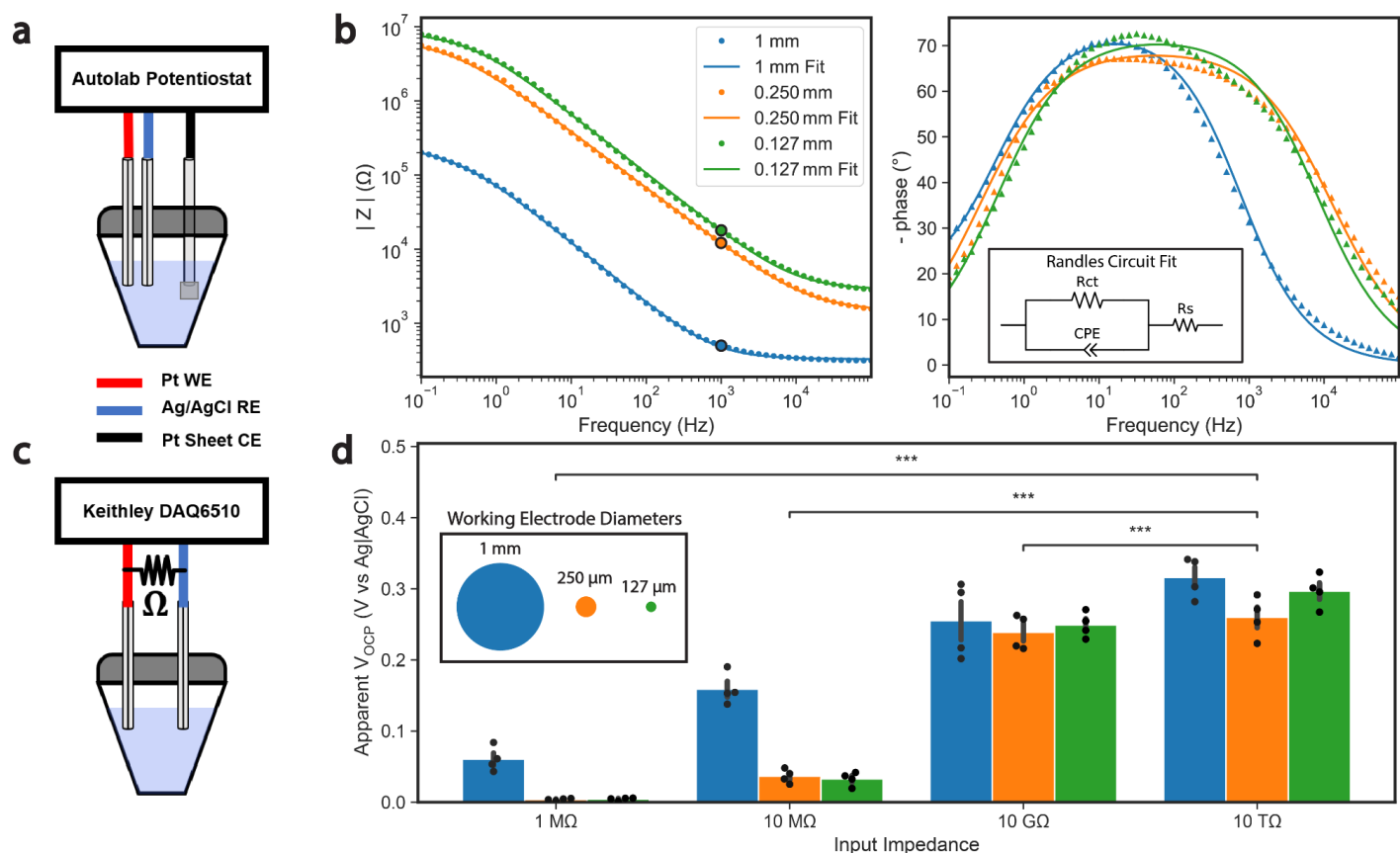


Figure 1: Characterization of DC electrode impedance (R_{dc}) and DC electrode voltage (V_{aOCP}) across input impedances. (a) Diagram of electrode configuration used to conduct EIS measurements. (b) Mean modeled vs experimental bode impedance magnitude and phase for electrodes of 1 mm (blue), 250 μm (orange) and 127 μm (green) diameter electrodes. Points outlined in black represent the impedance measured at 1 kHz. The Randles circuit diagram used to determine fit parameters from EIS spectra is present within the phase plot. (c) Diagram of electrode configuration used to conduct apparent open circuit potential (V_{aOCP}) measurements. (d) Apparent OCPs measured at different input impedances across electrode sizes. Legend in panel d is a scaled representation of the disk electrode diameters. All data in panel d are represented as mean $\pm$ SEM. *** for $p < 0.001$.

3.2 Impact of Input Impedance on Measured Electrode Potential

Equations 1 & 2 represent a simplistic theoretical calculation of expected V_{aOCP} by modeling the DC electrode impedance (R_{dc}) and measurement device impedance (Z_{in}) as a voltage divider, with equation 2 being the simplified form in terms of the ratio of R_{dc}/Z_{in} [35].

$$\frac{V_{aOCP}(\text{Measured})}{V_{aOCP}(\text{Actual})} = \frac{Z_{in}}{Z_{in} + R_{dc}} \quad (\text{eq. 1})$$

$$\frac{V_{aOCP}(\text{Measured})}{V_{aOCP}(\text{Actual})} = \frac{1}{1 + \frac{R_{dc}}{Z_{in}}} \quad (\text{eq. 2})$$

Figure 2(a) shows the relationship between increasing measurement device input impedance and R_{dc}/Z_{in} for electrodes of decreasing surface area. The measured apparent OCP values were normalized to the values collected at the 10 TΩ input impedance for each sample and plotted against the R_{dc}/Z_{in} (**Figure 2(b)**). The theoretical model defined by equations 1 and 2 is displayed as a black dashed line in **Figure 2(b)** and is overlaid with experimentally measured data to illustrate the relationship between the theoretical and observed V_{aOCP} values across varying R_{dc}/Z_{in} ratios. As can be seen, the functional measurements approximate the theoretical calculation

which generates a sigmoidal curve when plotted; however, this sigmoidal curve is shifted to the right for the functional measurements.

One possible source of this deviation of our functional measurements from theory is the method of EIS used to calculate the R_{dc} of each electrode. In our initial EIS measurements, we applied a sinusoid at each frequency centered on 0V vs Ag|AgCl. This is a common practice in neuromodulation for EIS measurements, because the OCP of an electrode is dependent on several factors (i.e. electrode surface state, concentration of redox species, etc.), creating a source of variability in the measurement of impedance at a specific frequency. Consequently, centering the sinusoids at 0V creates a common reference point mitigating this issue. However, applying a sinusoid centered at 0V forces the platinum electrodes away from their natural open circuit potential of $\sim 0.3V$. This requires inducing faradaic reactions, which in turn lowers R_{ct} .

To assess this possibility, we reconducted EIS with the applied sinusoids centered around the OCP of the electrodes; the OCP was measured for 60 seconds prior to the collection of each EIS spectrum. This should result in a much larger but more accurate estimate of R_{ct} (**Table 1**). **Figure 2(c)** displays the new ratios of R_{dc}/Z_{in} using this alternative approach. **Figure 2(d)** depicts the normalized OCP as a function of R_{dc}/Z_{in} calculated using these new EIS values versus the same functional data presented in **Figure 2(b)**. The new calculated expectation of normalized OCP as a function of R_{dc}/Z_{in} more closely approximates the measured functional data ($R^2 = 0.88$ @ OCP versus $R^2 = 0.44$ @ 0V vs Ag|AgCl), suggesting that performing EIS measures centered around the OCP of each electrode provided a more accurate representation of probe-loading effects.

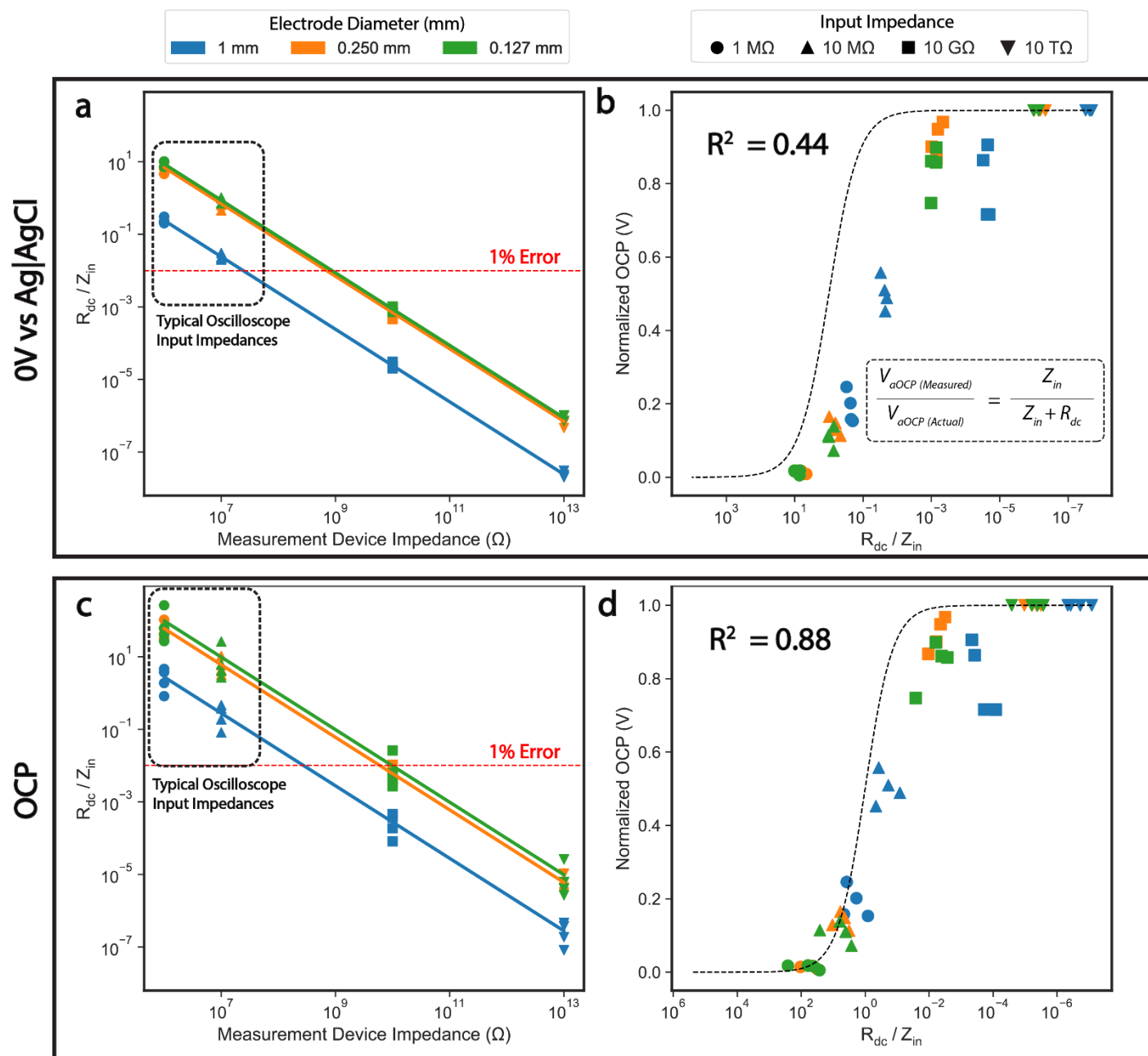


Figure 2: Significant measurement device backend impedance (Z_{in}) relative to DC electrode impedance (R_{dc}) is required to properly measure the DC electrode potential (V_{aOCP}) of an electrode. (a) and (c) Measurement device impedance vs R_{dc} / Z_{in} for each electrode where R_{dc} is derived from the Randles circuit fit and defined as $R_{ct} + R_s$ for Randles parameters taken from EIS performed around 0V vs Ag|AgCl and around OCP of each electrode, respectively. Dashed rectangle highlights typical input impedances for a standard oscilloscope probe. (b) and (d) R_{dc} / Z_{in} plotted against normalized apparent open circuit potential measurements. Black dashed line in both plots is the expected V_{aOCP} with the input impedance (Z_{in}) and the DC electrode impedance (R_{dc}) modeled as a voltage divider. Red dashed lines in panels a and c represent the ratio of R_{dc}/Z_{in} in which the idealized voltage divider model reaches a 1% measurement error.

3.3 Impact of R_{dc}/Z_{in} on Apparent Charge Injection Limit

Voltage transients (VT) are commonly used to determine the safe limits of stimulation for an implanted electrode. By measuring electrode polarization during therapeutic stimulation pulses, VTs help identify conditions that may drive unwanted faradaic reactions, leading to electrode dissolution and/or the generation of toxic

byproducts. For a platinum/platinum iridium electrode, a cathodic polarization of -0.6 volts or an anodic polarization of 0.8V relative to Ag|AgCl can cause the electrolysis of water, otherwise known as the ‘water window’ for platinum. This, in turn evolves H₂ or O₂, respectively, changing the pH of nearby tissue. Typically, the current/charge density just before these reactions occur is deemed the ‘charge injection limit’ (Q_{inj}) [17,32].

In the context of voltage transient responses, it is important to understand the distinct components of the voltage waveform and what each represents. **Figure 3(a) and (b)** depict the experimental set-up for measuring VTs, identifies these distinct components and depicts the process used in this study for determining the Q_{inj} .

In **Figure 3(b)**, V_{IPP} is defined as the interpulse potential, during which the stimulator current has been set to zero. In this study, V_{IPP} is used as a surrogate measure of DC electrode potential for our voltage transient experiments. Accurate measurement of the electrode’s interpulse potential (V_{IPP}) during voltage transient experiments is crucial for determining when the water window is breached and, consequently, for establishing the Q_{inj} of a given electrode material. The access voltage (V_a) arises from the instantaneous voltage drop across the electrolyte and is influenced by the electrolyte conductivity and cross-sectional area of the electrode. For biphasic stimulation waveforms, V_a appears as a step voltage change at the onset and termination of each phase due to the instantaneous change in current. V_a represents the voltage response due to the bulk impedance of the electrochemical system and is not associated with the electrochemical activity at the electrode interface; however, it is critical in the determination of the interfacial polarization voltage at the electrode.

In contrast, E_{mc} and E_{ma} represent the polarization experienced at the electrode interface in the cathodic and anodic direction, respectively. These values, relative to V_{IPP} , provide a direct measure of the capacitive and faradaic processes occurring at the interface and are used to evaluate whether the electrode remains within safe electrochemical limits. The Q_{inj} in this study is defined as the charge density in which the E_{mc} or E_{ma} exceed the cathodic or anodic limit for the water window (see **Figure 3(c)**). As VTs require the application of many pulses before reaching stability, particularly those conducted without DC mitigation, all calculations of Q_{inj} were performed after 750 biphasic pulses were applied which reached a point of stability for all electrode sizes and charge densities tested (see **Figure 3(d)**) [38].

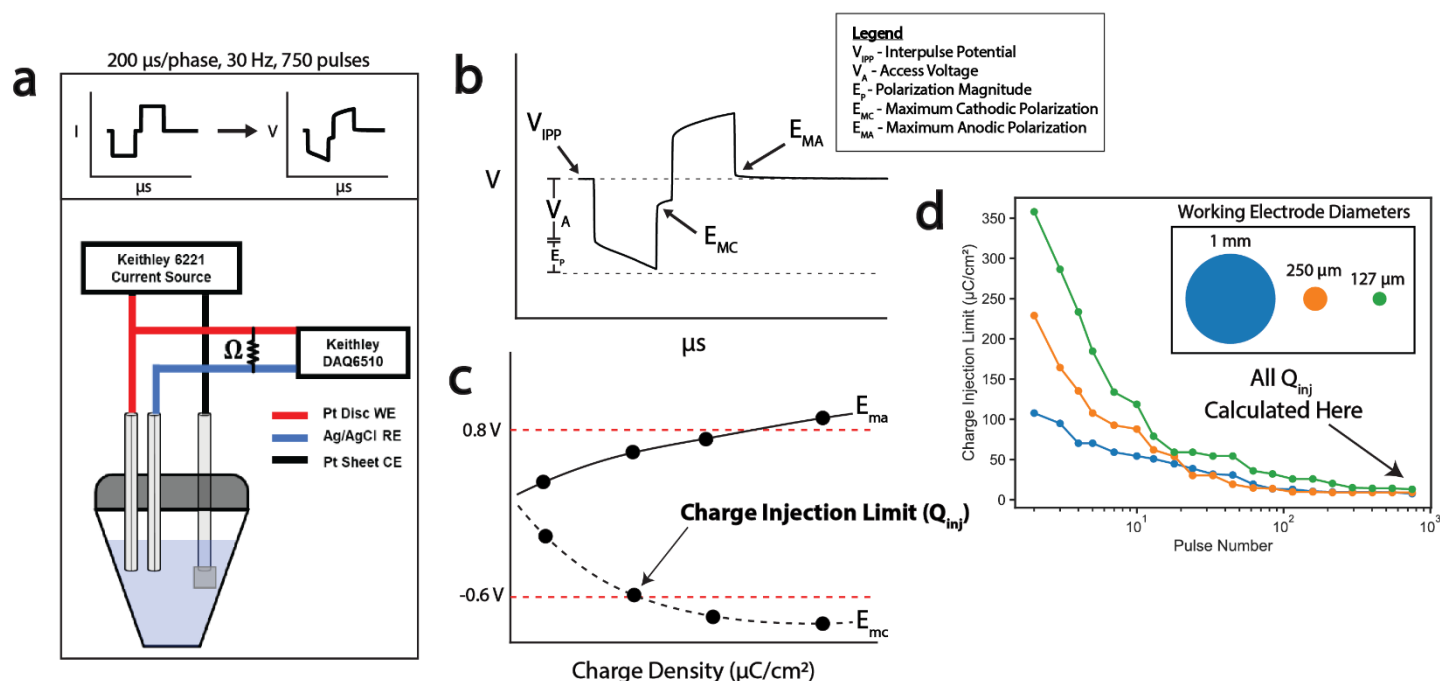


Figure 3: Experimental configuration and methodology used to determine charge injection limit (Q_{inj}). (a) Instrumentation configuration and pulse parameters for voltage transient measurements. The top panel depicts representative input current and output voltage traces. (b) Representative voltage transient with different waveform components highlighted. (c) Diagram of E_{mc} and E_{ma} plotted

as a function of charge density methodology for charge injection limit determination. (d) Theoretical charge injection limits calculated across pulses and sizes at 10 GΩ of input impedance.

Given that one of the goals of this study is to evaluate how instrumentation input impedance influences the measurement of the DC offset (ΔV_{IPP}), the data in this section are presented using a stimulation set-up anticipated to generate significant ΔV_{IPP} , i.e. stimulation with charge-balanced biphasic pulses without the use of a DC blocking capacitor or shorting of the cathode and anode between pulses. **Figure 4(a)** depicts a representative voltage transient for each electrode size obtained as a function of measurement device input impedance. The charge density of $\sim 100 \mu\text{C}/\text{cm}^2$ was selected as it is the midpoint of the range of 50-150 $\mu\text{C}/\text{cm}^2$, which has been identified as the Q_{inj} for Pt/Ir electrodes in prior studies. The largest ratio of electrode impedance to instrumentation impedance is at DC (i.e. R_{dc}/Z_{in}) thus V_{IPP} is anticipated to be the voltage transient component that is most impacted by changes in measurement device input impedance. Conversely, V_a is a very high frequency component of the VT. At this frequency the electrode impedance is dominated by R_s which is orders of magnitude lower than R_{dc} . Consequently, the ratio of instrumentation impedance (Z_{in}) to the high frequency electrode impedance (R_s) for the measurement of V_a is orders of magnitude larger, therefore the measurement of V_a is anticipated to be minimally impacted by changes in measurement device input impedance.

Consistent with theoretical expectations, as a measure of DC electrode potential, the measurement of V_{IPP} during VTs at each electrode size, shown in **Figure 4(b)**, follows the same pattern as our prior measurements of V_{aOCP} . The measured V_{IPP} of the largest electrode size of 1 mm diameter which has the lowest DC impedance is the most consistent across the input impedances tested (an average of $1.21 \pm 0.885 \times 10^{-2}$, $1.28 \pm 1.87 \times 10^{-2}$, $1.27 \pm 1.89 \times 10^{-2}$ and $1.31 \pm 1.73 \times 10^{-2}$ V at 1 MΩ, 10 MΩ, 10 GΩ, and 10 TΩ, respectively). Conversely, the smallest electrode size of 0.127 mm diameter with a much higher DC impedance is most impacted by the input impedance of the measurement system (averaging $8.58 \times 10^{-2} \pm 1.45 \times 10^{-3}$, $7.47 \times 10^{-1} \pm 1.65 \times 10^{-2}$, $1.08 \pm 0.510 \times 10^{-2}$, and $1.02 \pm 4.14 \times 10^{-3}$ V at 1 MΩ, 10 MΩ, 10 GΩ, and 10 TΩ, respectively). Importantly, as the input impedance of the measurement device increases, the measured V_{IPP} at each electrode size converges to between 1.1 and 1.3V even at much greater charge densities than 100 $\mu\text{C}/\text{cm}^2$ (see **Figure 4(b)**).

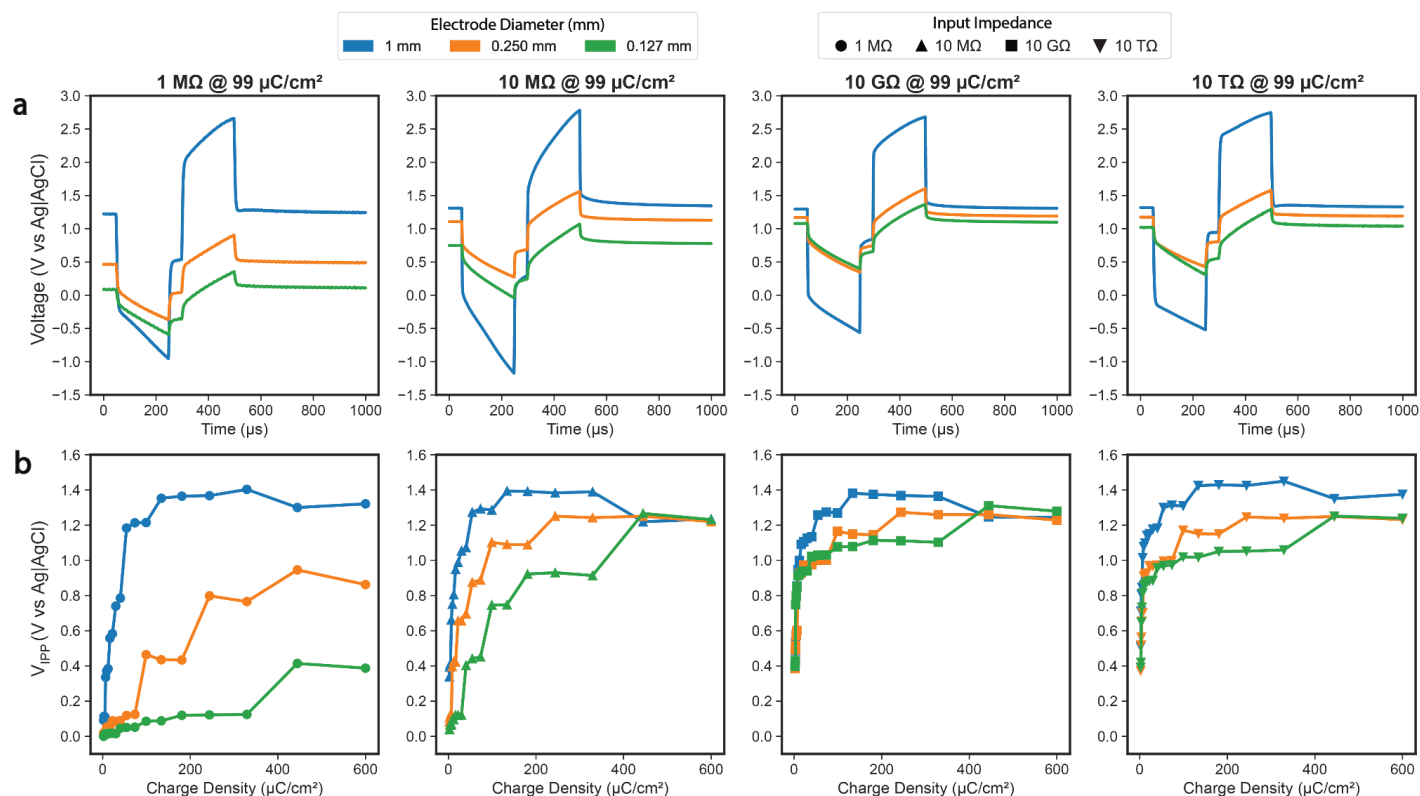


Figure 4: Observed changes in representative voltage transients and observed interpulse potential (V_{IPP}) across input impedances. (a) Average representative voltage transients across sizes collected at $\sim 100 \mu\text{C}/\text{cm}^2$ across input impedances and electrode diameters. (b) Average V_{IPP} across input impedances and electrode diameters. Depicts the VT waveform shape and observed convergence of V_{IPP} at larger input impedances across increasingly smaller electrodes.

As expected, the measurements of V_a are relatively consistent across measurement system input impedances. To maintain a constant charge density across electrode sizes, the applied current was scaled proportionally to the electrode surface area. Because the applied current scales with the square of the radius, the current delivered to the 1 mm diameter electrode was ~ 61 times greater than that applied to the 0.127 mm diameter electrode. According to Ohm's law ($V_a = I \times R_s$) and based on the geometric relationship between R_s and electrode radius for circular electrodes described in equation 3, where R_s is the solution resistance, ρ is the electrolyte resistivity and r is the electrode radius. This difference in current and R_s results in an expected $\sim 8x$ larger V_a for the 1 mm diameter electrode compared to the 0.127 mm diameter electrode [33–35]. Across measurement input impedances of 1 MΩ, 10 MΩ, 10 GΩ and 10 TΩ, the average V_a measured during the VT for the 0.127 mm diameter electrode was $1.65 \times 10^{-1} \pm 3.03 \times 10^{-3}$, $2.09 \times 10^{-1} \pm 1.11 \times 10^{-2}$, $2.04 \times 10^{-1} \pm 6.31 \times 10^{-3}$, and $1.88 \times 10^{-1} \pm 3.28 \times 10^{-3}$ V, respectively. By comparison, the measured V_a during the VT for the 1 mm diameter electrode are $\sim 6-8x$ larger; $1.27 \pm 6.01 \times 10^{-2}$, $1.32 \pm 2.96 \times 10^{-2}$, $1.32 \pm 3.20 \times 10^{-2}$ and $1.34 \pm 1.08 \times 10^{-1}$ V for the 1 MΩ, 10 MΩ, 10 GΩ, and 10 TΩ measurement input impedances, respectively. The V_a for the 0.250 mm diameter electrodes lies in between the 0.127 and 1mm diameter electrodes, and are approximately 2x the 0.127 mm V_a , matching theoretical expectations.

$$R_s = \frac{\rho}{4r} \quad (eq. 3)$$

In comparison to V_{IPP} and V_a measurements during VTs, which closely match theoretical expectations, the measurements of cathodic and anodic polarization on the electrodes across input impedances are much more complex. This is not unexpected, as in addition to the polarization at the electrode interface having both high

frequency and low frequency components, it is also heavily dependent on the surface state of the electrode [29,39,40].

Figure 5(a) shows the impact of the larger observed ΔV_{IPP} highlighted in **Figure 4** on calculated values of E_{mc} and E_{ma} which are used to determine Q_{inj} . **Figure 5(b)** shows the mean representative voltage transients collected at the apparent Q_{inj} across electrodes of decreasing surface area without DC mitigation, which can occur at different current densities depending on the electrode size and input impedance of the VT measurement system. At 1 M Ω of input impedance, the apparent Q_{inj} is disproportionately higher for smaller electrodes (215 ± 3.31 , $250 \pm 15.6 \mu C/cm^2$ for 0.250 mm and 0.127 mm diameters, respectively), and the polarization direction leading to the potential limit breach varies by electrode size – E_{ma} for the 1mm and 0.250 mm diameter electrodes versus E_{mc} for the 0.127 mm electrodes. As input impedance increases, the breach consistently occurs at E_{ma} across all electrode sizes and at much lower charge densities. This pattern underscores how measurement errors driven by a high R_{dc}/Z_{in} ratio can lead to an overestimation of the Q_{inj} of platinum electrodes due to an underestimation of ΔV_{IPP} at the electrode interface which is depicted in **Figure 5(c)**.

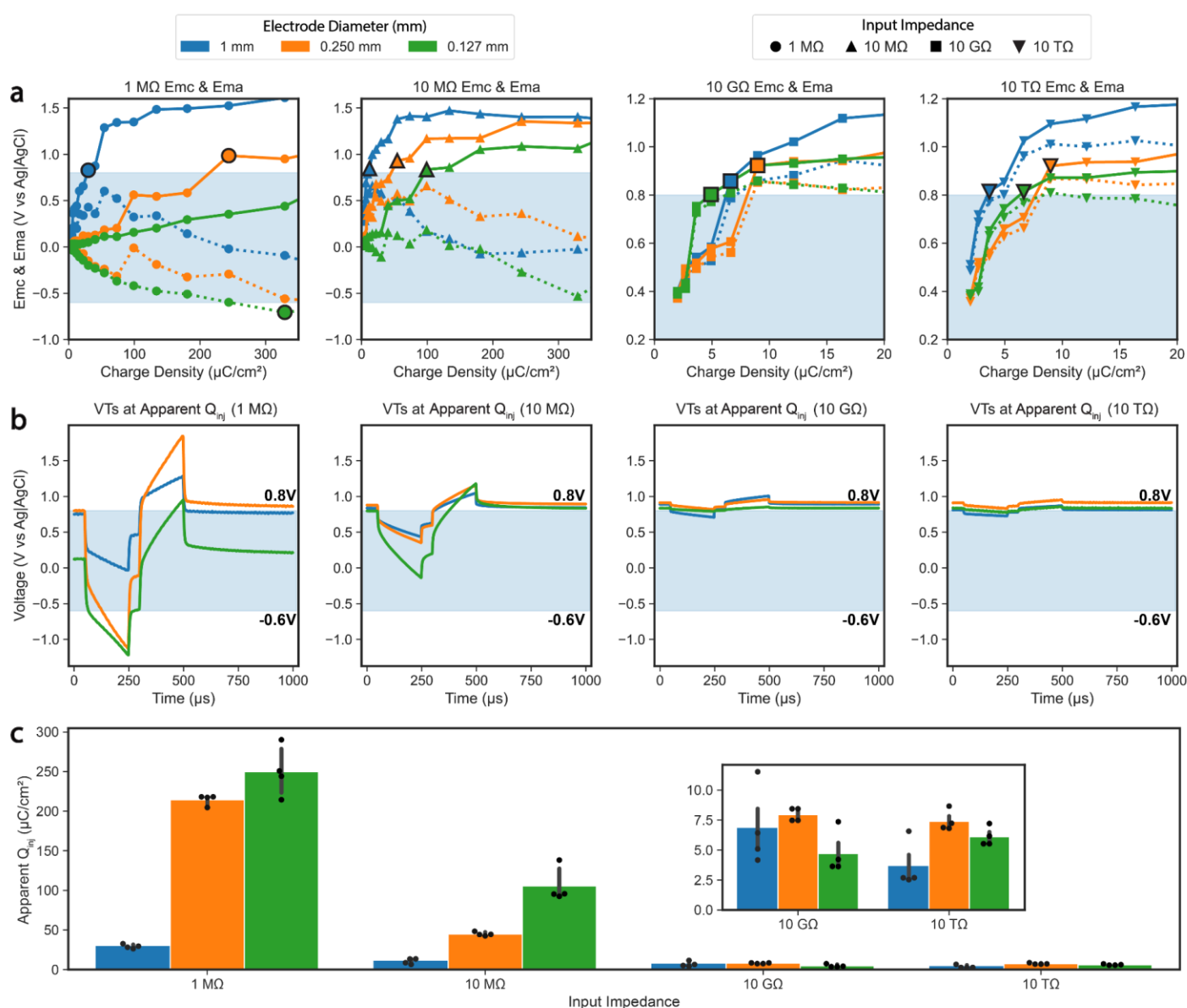


Figure 5: Traditional methods for calculating charge injection limits (Q_{inj}) are susceptible to changes in the ratio of DC electrode impedance to the measurement device input impedance (R_{dc}/Z_{in}). (a) Maximum anodic and cathodic polarization (E_{mc} , E_{ma})

calculated for the last pulse across input impedances and sizes. The markers with black edges represent the point in which either E_{mc} or E_{ma} crossed the water window which is marked in panels a & b by the blue shaded region. (b) Representative mean voltage transient response of the last pulse applied at the charge injection limit (Q_{inj}) across sizes. (c) Corresponding Q_{inj} calculated from data shown in b. All data in panel c are represented as mean $\pm$ SEM.

3.4 Impact of DC Mitigation Strategy on Apparent Charge Injection Limit

The two most common methods used to mitigate stimulator leakage currents during stimulation with imbalanced or balanced biphasic pulses are the use of an inline capacitor (CC) and/or the use of an inline capacitor while shorting the cathode/anode in between pulses (CC+ES) [11,18,41–45]. The impedance of a capacitor is inversely proportional to frequency, such that an ideal capacitor functionally has an infinite impedance at DC. Consequently, the inline ‘blocking’ capacitor is used to block stimulator leakage currents from reaching the electrodes. However, a capacitor allows current flow until the voltage across the capacitor equals the voltage of the source (i.e. the stimulator). As shown in **Figure 6(b)** below, when just an inline capacitor is used without shorting the cathode and anode in between pulses, a growing DC offset at the working electrode is still evident over the course of the first ~400 pulses (See **Figure 6(b)**, 1st, 2nd panel; **Figure 6(c)**). Simultaneously, the voltage across the counter electrode blocking capacitor increases linearly due to the unmitigated leakage current of the Keithley current source (**Figure 6(d)**). Around pulse 425, the voltage across the capacitor plateaus and one phase of the biphasic waveform is increasingly ‘cut off’, colloquially known as ‘clipping’ or ‘railing’, resulting in biphasic current waveforms that are no longer charge balanced (See **Figure 6(b)**, 3rd and 4th panel; **Figure 6(d)**). This phenomenon occurs because the voltage of the system (i.e. the sum of the voltages across both blocking capacitors, and both the working and counter electrodes) has reached the 20V compliance limit of the stimulator. Note, at this point V_{IPP} begins to decrease, ultimately settling at a new equilibrium point around ~0.6V (See **Figure 6(d)**, 4th panel) putatively due to the compromised charge balance of the biphasic current waveform counteracting the stimulator leakage current.

For this reason, many clinical implantable pulse generators (IPGs) use both an inline capacitor and short behind the electrodes in between pulses, to bleed off charge that has accumulated on the capacitor and prevent clipping/railing of the biphasic pulse. Unfortunately, many stimulators used for preclinical animal research and benchtop testing of electrodes either lack a DC leakage current mitigation strategy or, if capacitively coupled, do not short the cathode/anode between biphasic pulses. This is exacerbated by the fact that measurement systems with insufficient input impedances are often used to measure the output waveform, which as shown previously can dramatically underestimate changes in V_{IPP} .

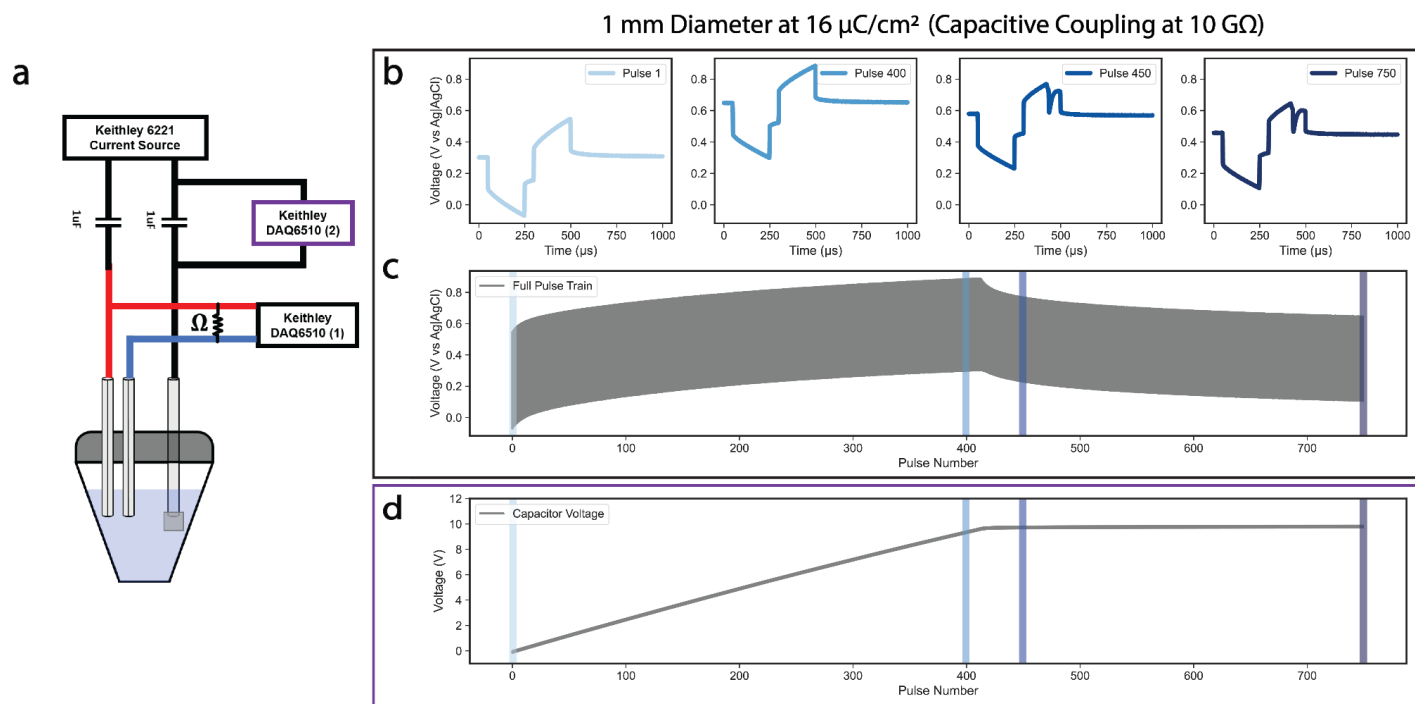


Figure 6: Capacitive coupling alone does not natively improve charge balance of constant current stimulators. (A) Specific experimental configuration used to collect the data displayed in panels b-d. (B) Representative voltage transient pulses (1, 400, 450, 750) collected at a 1 mm diameter electrode over the course of a pulse train at $\sim 16 \mu\text{C}/\text{cm}^2$ during the capacitive coupling (CC) condition. (C) Raw voltage plotted against pulse number with colored regions illustrating regions of the pulse train that are represented in panel a. (D) Voltage across the counter electrode blocking capacitor as a function of pulse number.

Figure 7(a) and 7(b) contrasts the lack of a DC mitigation strategy with CC+ES on a representative example of a single voltage transient trial at $54 \mu\text{C}/\text{cm}^2$ with 10 G Ω of input impedance. In this example, all 750 voltage transient pulses for each condition are plotted on the same axis for each electrode size. The corresponding insets represent the ΔV_{IPP} observed over the course of the pulse train. It is evident from these plots that a significant reduction in ΔV_{IPP} was observed with CC+ES. **Figure 8** further elaborates on this observation by comparing the average ΔV_{IPP} as a function of the charge densities applied to each electrode with and without proper DC mitigation. Across the charge densities tested, CC+ES resulted in a reduction of average $\Delta V_{\text{IPP}} \geq 88\%$ as measured at an appropriate input impedance.

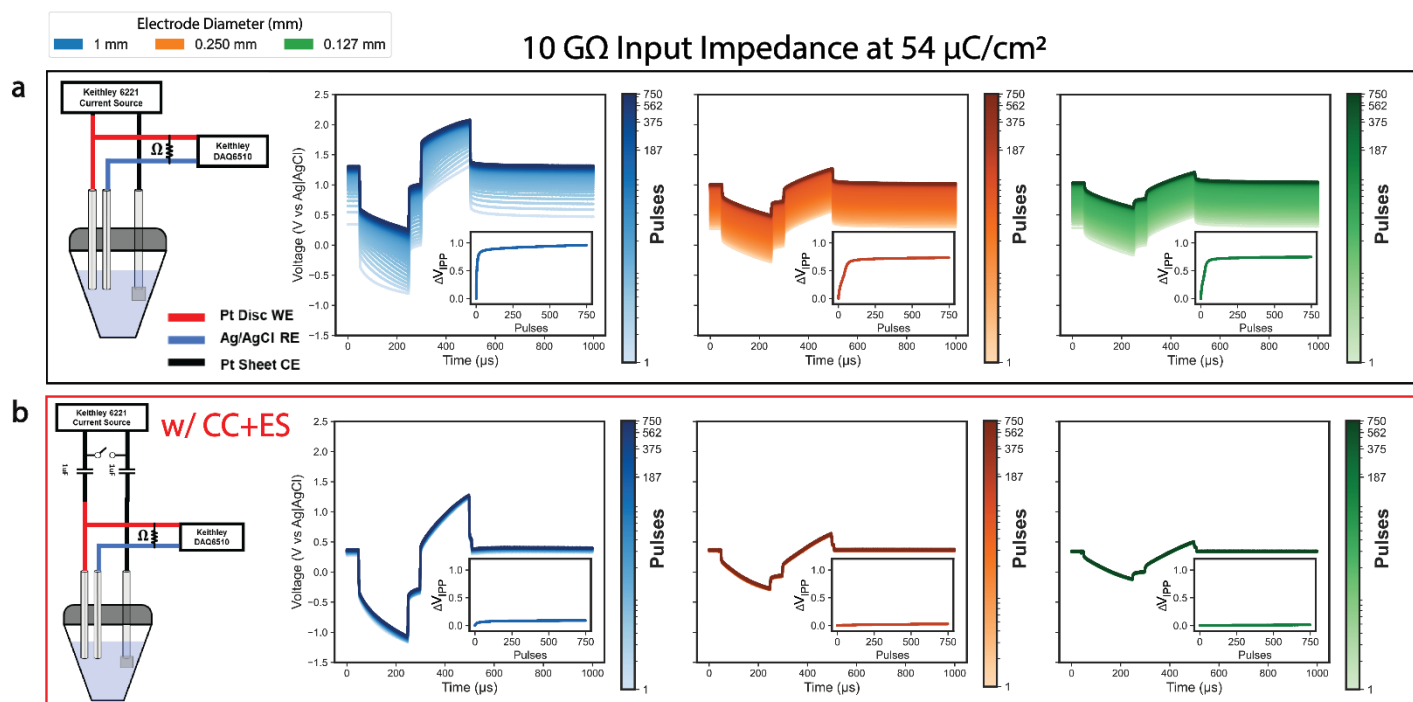


Figure 7: Representative examples showing that capacitive coupling with electrode shorting (CC+ES) greatly reduces the DC offset (ΔV_{IPP}) observed during voltage transients. Representative voltage transients collected at $54 \mu\text{C}/\text{cm}^2$ and $10 \text{ G}\Omega$ of input impedance with (a) and without (b) custom DC mitigation strategy across sizes. Insets of each plot represent the change in ΔV_{IPP} observed for that trial.

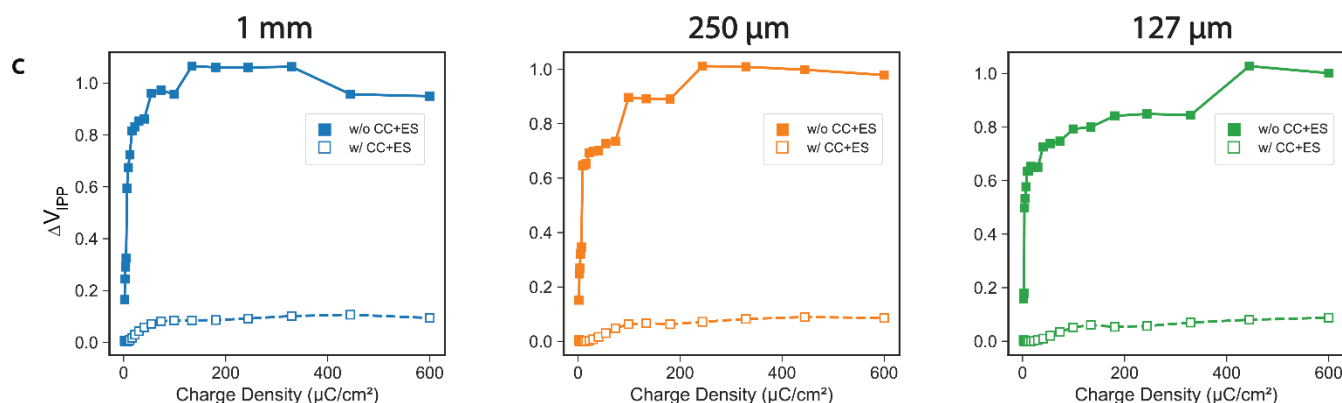


Figure 8: Capacitive coupling with electrode shorting (CC+ES) greatly reduces the DC offset (ΔV_{IPP}) observed during voltage transients across charge densities. Average ΔV_{IPP} vs charge density collected with and without custom DC mitigation strategy at $10 \text{ G}\Omega$ of input impedance across sizes. Note, the ΔV_{IPP} plots for the trials without DC mitigation are transformed from the data presented in **Figure 4(B)** at $10 \text{ G}\Omega$.

As discussed in **Section 3.3**, the VT measurements performed with sufficient input impedance and without DC mitigation greatly reduced the calculated Q_{inj} at platinum electrodes using traditional methods. This reduction is due to unmitigated stimulator leakage currents leading to larger observed changes in V_{IPP} . However, when these leakage currents are properly mitigated, this alters Q_{inj} . **Figure 9(a)** plots the Q_{inj} as a function of R_{dc}/Z_{in} across the input impedances, electrode sizes and DC mitigation strategies tested while also contextualizing these results in reference to the historical $50\text{-}150 \mu\text{C}/\text{cm}^2$ limits which are shaded in green. Note, the low input impedance, DC mitigated trials consistently produced Q_{inj} that were in line with the values derived by Robblee, et al. Without proper DC mitigation, as R_{dc}/Z_{in} decreases, so does the calculated Q_{inj} (i.e. 250 ± 15.6 , 105 ± 11.0 , and $4.71 \pm$

0.894 $\mu\text{C}/\text{cm}^2$ at 1 M Ω , 10 M Ω , 10 G Ω , respectively for the 0.127 mm diameter electrodes); however, in this study, an inverse relationship was observed with proper DC mitigation (i.e. 155 ± 44.1 , 243 ± 23.0 , 359 ± 13.8 $\mu\text{C}/\text{cm}^2$ at 1 M Ω , 10 M Ω , 10 G Ω , respectively for the 0.127 mm diameter electrodes). This suggests that with appropriate measurement instrumentation in addition to a properly charge balanced stimulator, appreciably larger charge injection limits can be safely applied beyond what has been reported historically [41]. **Figure 9(b)** represents an alternative grouping of these results that demonstrates the average effect of DC mitigation on the calculated Q_{inj} at different input impedances. There were significant differences in Q_{inj} across input impedances both with and without CC+ES (Friedman: $\chi^2 = 17.17$ and 22.17 ; $p < 0.001$ and $p < 0.0001$; Kendall's $W = 0.72$ and 0.92 , respectively). Post-hoc Wilcoxon signed-rank tests showed significant pairwise differences in Q_{inj} when comparing each input impedance to the 10 G Ω condition (Holm-corrected $p < 0.01$, effect size $r > 0.79$). Additionally, there was a statistically significant difference in Q_{inj} with and without CC+ES (Wilcoxon signed-rank test: $p < 0.0001$, effect size $r = 0.81$). ***These data stress that both the ratio of the DC electrode impedance to the measurement device input impedance and the DC mitigation strategy deployed have a significant impact on Q_{inj} .***

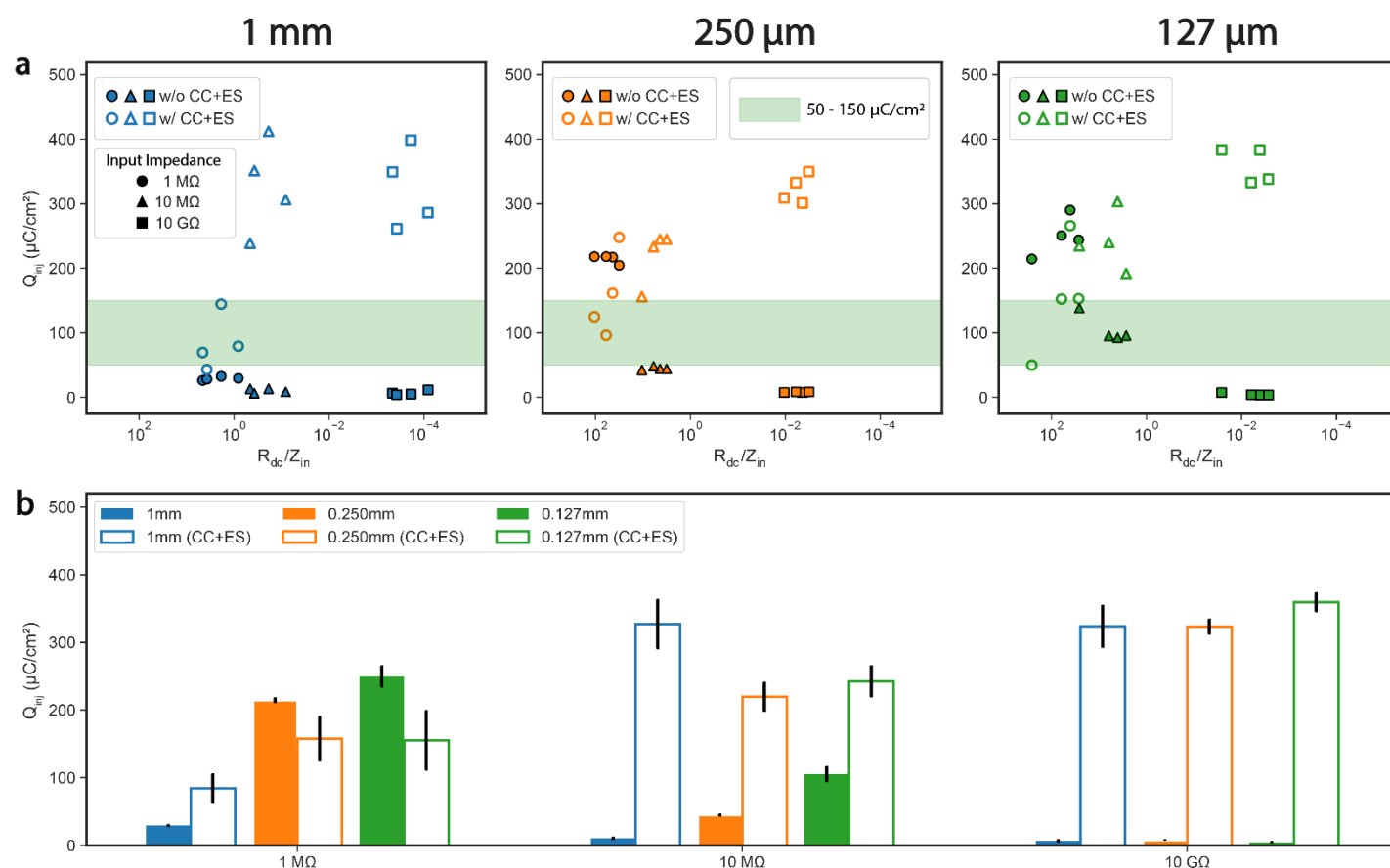


Figure 9: Experimentally derived charge injection limits (Q_{inj}) can differ widely depending on both $R_{\text{dc}}/Z_{\text{in}}$ and DC mitigation strategy. (a) $R_{\text{dc}}/Z_{\text{in}}$ (as determined by EIS around the open circuit potential (OCP)) vs charge injection limit for each sample across sizes, input impedances with and without DC mitigation. The highlighted region in green represents the 50-150 $\mu\text{C}/\text{cm}^2$ limits determined by Robblee, 1990 [12]. (b) Charge injection limits across sizes, with and without DC mitigation, grouped by input impedance. Solid bars in panel b are identical to the charge injection limits represented in **Figure 5(c)**. All data in panel b are represented as mean $\pm$ SEM.

4. Discussion

To safely stimulate neural tissue, charge can be delivered through either capacitive or faradaic charge transfer mechanisms. Capacitive charge transfer is generally considered the safest way to inject charge for neural stimulation. Conversely, charge passed through faradaic mechanisms can be unsafe if the reactions are not completely reversed and form toxic byproducts or corrode the electrode. In practice, all electrodes have a limited amount of charge that can be passed exclusively through safe reactions. A fundamental and unresolved challenge in designing stimulation electrodes is determining stimulation limits that enable sufficient charge injection to achieve therapeutic effect but prevent reactions that are unsafe to tissue or damage the electrode.

Voltage transients (VTs) have been commonly used to assess what reactions are available for faradaic charge transfer and how much current can be passed before the onset of unsafe reactions. In this paper, we demonstrate that the accuracy of voltage transient measurements can be impacted by stimulation and measurement set-ups commonly used in literature. These data stress the importance of characterizing both the stimulator set-up as well as the measurement system to enable consistent and accurate determination of Q_{inj} from VTs. Based on these results, we comment on the importance of mitigating leakage currents at neural stimulators and suggest reporting strategies to enable reproducible VTs for characterization of materials, development of neural stimulators, and evaluation of complete neuromodulation systems.

4.1 Measuring the Electrochemistry of Neural Stimulation

The amount of current that can be passed through capacitive mechanisms primarily depends on the size and surface morphology of the electrode. In contrast, the tolerable amount of faradaic charge transfer depends on availability of faradaic reactions that are either highly reversible or are irreversible but result in safe reaction products. Although a simplification, the availability of these reactions could be considered to depend on four properties: 1) the electrode material 2) the electrochemically active surface area 3) the concentration of reactants/products for faradaic reactions in the solution around the electrode, and 4) the electrochemical state of the electrode surface, which impacts the available faradaic reactions. While the first 2 of these properties are determined by the electrode being tested, the surface state of the electrode and concentration of reactants/products around the electrode must be controlled to maintain consistent charge injection capabilities.

The potential of an electrode in solution, measured at DC, is the mixed Nernst potential resulting from the balance of several half reactions within the electrolyte. The mixed Nernst potential depends on the available faradaic reactions at the electrode and the concentration of reactants/products needed for those reactions. A change in the electrode potential, when no net current is being passed, can be indicative of a change in the faradaic reactions available for charge injection, either by changes in the surface state of the electrode (i.e. PtO formation) or the concentrations of redox species in the electrolyte. Our data demonstrates that measurement and stimulation systems can interact with an electrode, changing the measured DC electrode potential, and thus influence the Q_{inj} measured via VTs.

4.1.1 Effect of Measurement Instrumentation on DC Electrode Potential

An ideal voltage measurement would have infinite input impedance, resulting in an accurate measurement of the DC electrode potential that isn't subject to probe loading effects introduced by the measurement device [46]. In practice, oscilloscopes, typically with input impedances of 1-10 M Ω s, are commonly used to measure electrode voltage while conducting VTs. This is much lower than the >10 T Ω input impedance for common electrochemical instrumentation (i.e. potentiostats). At oscilloscope input impedances, we observed a significant difference in the measured DC electrode potential (V_{aOCP}) from historically reported values across all electrode

sizes (**Figure 1(d)**). This observation highlights a clear dependence on the relationship between the ratio of R_{dc}/Z_{in} and measured DC electrode potential. Smaller electrodes, which have a higher R_{dc} , showed the greatest deviation from the expected DC electrode potential (**Figure 1(d)**). In other words, oscilloscope input impedances are often insufficient even for measuring the DC potential of Pt macroelectrodes (defined as geometric surface area greater than $2 \times 10^{-5} \text{ cm}^2$) and the error becomes dramatically worse as electrode size decreases. This measurement error can then propagate, leading to misrepresentations in experimental determinations of Q_{inj} . This presents a critical confound that must be considered to improve rigor and reproducibility of in-vivo and in-vitro measures of neuromodulation device safety and longevity, especially with the increasing popularity of TFMEAs.

4.1.2 Appropriate Measurement Instrumentation is Essential for Accurate Q_{inj} Determination

To ensure that the Q_{inj} determined from voltage transients are valid, it is best to accurately measure the DC electrode potential (V_{IPP}) relative to a stable electrochemical reference. When the input impedance of the measurement system is too low relative to the electrode's DC impedance, changes in V_{IPP} are poorly resolved, leading to incorrect assessments of when the electrode reaches the limits of water electrolysis. This undermines the core premise of voltage transient analysis, which is to infer the onset of irreversible faradaic processes from the observed electrode interfacial polarization. Without proper tracking of V_{IPP} , particularly in two-electrode configurations or systems lacking sufficient measurement device input impedance, it becomes impossible to accurately determine the electrode's potential relative to the desired electrochemical window. As a result, it may not be possible to distinguish between reversible and irreversible charge transfer mechanisms. As such, accurate measurement and reporting of V_{IPP} is a prerequisite for determining parameters of safe stimulation across studies [33,39,40]. We recognize that this issue is difficult to rectify during chronic in-vivo studies, as maintaining a stable reference electrode within tissue chronically is extremely difficult for a multitude of reasons (i.e. biofouling, Ag toxicity). To overcome these challenges, future studies should explore more stable reference electrode materials, more robust and rigorously validated DC mitigation strategies or the development of algorithms that accurately predict DC potential shifts during 2-electrode measurements.

4.2 DC Electrode Potential as an Indicator of Stability during Voltage Transients

As previously described, the DC potential of an electrode reflects the balance of available reactants, reaction products and the electrochemical state of the surface. For Pt electrodes, these factors define distinct polarization regions in which different faradaic pathways dominate, each with varying charge transfer capabilities. In our data, V_{IPP} – charge density curves consistently exhibited plateaus near 0.3V, 0.5V and 0.8V vs Ag|AgCl, corresponding to potential regions associated with oxygen reduction, oxide/chloride formation and oxygen evolution, respectively [29]. These plateaus denote regions where the electrode interface can temporarily buffer additional charge through specific faradaic reactions; once the buffering capacity of that potential region is exceeded, V_{IPP} rises sharply as the electrode enters the next dominant redox pathway.

Although electrochemical safety is often determined by electrode polarizations leading to the electrolysis of water (i.e. -0.6V and +0.8V vs Ag|AgCl), at lower levels of polarization, other reactions occur that can lead to the loss of electrode material and/or the formation of toxic byproducts. For example, the formation of platinum chloride begins at ~0.5V vs Ag|AgCl, well within the nominal water window, and has been linked to accelerated Pt dissolution and neural toxicity [47–49]. When DC offsets are accurately measured and stimulator leakage currents are successfully minimized, these offsets provide a sensitive indicator of the onset of such degradation mechanisms [29].

To disentangle instrumentation-driven effects from true electrochemical behavior we applied the same stimulation amplitudes and DC mitigation strategies to a simple dummy load (**Supplementary Figure 5a**). These measurements serve two purposes: 1) To validate the impact of the DC mitigation strategies on the stimulator leakage current and 2) To demonstrate the differences in stimulator leakage current specifications across common preclinical stimulators (Keithley 6221 vs TDT Subject Interface) despite delivering the same charge-balanced waveform (**Supplemental Figure 5b-e**). In other words, “charge-balanced” pulses are not equivalent across devices and instrumentation alone can introduce non-trivial differences in net delivered charge.

In contrast to the dummy load, which showed essentially zero measurable leakage current when using our mitigation strategy, the in-vitro recordings still exhibited small but consistent shifts in V_{IPP} (**Figure 8**). However, these offsets cannot be attributed to the stimulator and instead arise from the electrochemical interface itself. Their presence, even under truly charge-balanced stimulation, indicates that the faradaic reactions supporting charge transfer are not perfectly reversible. As a result, irreversible reaction products accumulate and perturb the local electrode environment over time, producing a residual DC offset that reflects the underlying reaction asymmetry. This distinction between charge balance (equally injected/withdrawn current) and electrochemical balance (no net chemical change at the interface) is critical [17]. Different electrode materials rely on unique faradaic pathways to support charge transfer, which inherently differ in their reaction kinetics. Several studies report that deliberately ‘breaking’ charge balance, via small biases or asymmetric pulses, can reduce dissolution by driving the interface toward a more electrochemically favorable operating point [50–52].

Taken together, our findings support the growing view that a properly measured DC offset, once the instrumentation-related confounds are rectified, provides a meaningful readout of the electrochemical stability at the electrode interface [19]. More importantly, they highlight material-specific waveform optimization as a potential strategy to minimize electrode dissolution and improve the longevity of neuromodulation devices.

4.3 Validating DC Mitigation Strategies with Proper Instrumentation

The use of DC blocking capacitors on stimulator output stages is considered best practice both clinically and preclinically to prevent prolonged DC accumulation at an electrode. However, as demonstrated in **Section 3.4**, simply adding capacitors does not result in a charge balanced stimulator that effectively eliminates ΔV_{IPP} . In fact, infrequent discharge of the DC blocking capacitors can result in decreased stimulator voltage compliance, or the accumulation of significant and potentially harmful offset voltages [53,54]. As noted in this study, once the stimulator reaches compliance, waveform asymmetry and truncation can occur, undermining charge balance and leading to unpredictable or undesirable electrochemical behavior at the electrode-electrolyte interface. (**Figure 6**). It is also worth noting that successful DC mitigation via DC blocking capacitors and electrode shorting, as shown in this study, requires careful selection of components and circuit design to minimize parasitic resistances. This is necessary to ensure all charge accumulated on the DC blocking capacitors during a stimulation pulse is discharged back through the stimulation electrodes between pulses and not lost to parasitic circuit pathways.

The DC mitigation strategy proposed in this study required the use of large DC blocking capacitors on the stimulator output. This approach becomes impractical with multichannel, high density implantable devices. As a result, several alternative DC mitigation strategies that have been proposed in recent years to reduce their physical footprint and cost [55–57]. To rigorously validate the effectiveness of these alternative strategies, it is imperative to use appropriate measurement device input impedances to properly measure DC accumulation, particularly for TFMEAs due to their susceptibility to corrosion-related failures and higher impedance.

4.4 Avoiding Confounds: Practical Experimental Considerations and Reporting Strategies

Rigorous reporting of the specifications of the instrumentation (Z_{in} , stimulator leakage currents, DC mitigation strategies) used to determine the Q_{inj} for a particular electrode is critical for the interpretation of electrochemical test results across studies and materials [33]. In every study reporting Q_{inj} for neuromodulation electrode materials, the DC electrode impedance (R_{dc}), and relevant stimulator/measurement device specifications (i.e. leakage current, input impedance) should also be disclosed. This data is necessary to evaluate the influence of instrumentation on observed V_{IPP} .

It is important to emphasize that the results reported in this study are specific to the combination of instrumentation, experimental parameters, electrode materials and DC mitigations strategies used. Changes to any component of the system may alter experimental outcomes. Proper contextualization of the overall system used to obtain results is crucial in the comparison of safety limits across labs and studies.

4.4.1 Practical Considerations and Implications for Neural Stimulation in Neuromodulation Experiments

In vitro charge injection limits are often used to justify that the stimulation parameters applied during in-vivo experiments fall within electrochemically safe boundaries. Our findings suggest that in vivo results may be biased if the relationship between the stimulator's leakage currents and DC mitigation strategy is not properly characterized. Inadequate DC mitigation can lead to unintended charge accumulation at the electrode interface, resulting in local pH shifts, sustained DC electric fields or other effects that may alter neuronal excitability and tissue health [21,24,25,58,59]. Additionally, these issues can cause unpredictable stimulator behavior, such as waveform truncation, which may go unnoticed but significantly bias functional outcomes. These consequences are particularly relevant for small electrodes, like those commonly used in preclinical rodent models, where small stimulator leakage currents can result in large shifts in electrode potential. Consequently, there may be cases within the literature where Q_{inj} and, by extension, the safety margins of a given neuromodulation system or material, are a misrepresentation of the true functional and electrochemical performance. Future studies should take care to ensure stimulator leakage currents are appropriately mitigated for a given electrode material and consider that oscilloscope measurements may be misleading (**Table 2**). Additionally, future work should critically assess how instrumentation-related errors may have influenced historically reported functional and safety related outcomes.

4.4.2 Practical Considerations for Reproducible and Consistent Assessment of Electrode Materials using Voltage Transients

Preventing corrosion and reaction byproduct accumulation are crucial to ensuring the longevity and safety of neuromodulation devices. This presents a larger issue at microelectrodes versus macroelectrodes due to smaller margin of error before critical failure, especially for chronically implanted active electrodes. The data presented herein provide a plausible explanation as to why premature failure of TFMEAs is so common. Our data also have consequences related to the validation and characterization of novel electrode materials. While platinum was the only material characterized in this study, the results are relevant and can be extrapolated to other electrode materials. For example, electrodes and electrode coatings that are faradaic in nature (i.e. IrOx) will have a lower R_{ct} , meaning that these electrodes will need a lower Z_{in} to properly measure the DC electrode potential. In contrast, electrode materials and coatings that are capacitive in nature (i.e. TiN) will have an inherently higher R_{ct} requiring a higher Z_{in} . Electrode surface area and fractal geometry will also influence these electrode parameters. A faradaic electrode with a larger electroactive surface area will ultimately lower R_{ct} and thus reduce R_{dc}/Z_{in} [60,61]. Studies

characterizing novel electrode designs and materials for stimulation should carefully evaluate if their instrumentation provides an accurate measurement of the Q_{inj} of the material. To do this, researchers should report the parameters outlined in **Table 2** depending on specific experimental goals.

4.5 On the Utility of Voltage Transients in Assessing Electrode Reactions and Safety

Neural electrode safety is a multifactorial problem that cannot be completely assessed via in-vitro performance tests. In-vitro tests like voltage transients allow for rapid comparisons between devices to be readily made, but as demonstrated by our results, these comparisons must be contextualized in reference to the instrumentation used [62]. It is also important to note that in-vitro tests are not a measure of safety alone but give an idea of what types of electrochemical reactions may occur at the electrode interface during a particular therapy. There are several confounds that exist in the in-vivo electrode environment due to its dynamic nature that are difficult to replicate in-vitro. Aspects like the protein adsorption and the foreign body response can drastically change the electrode interface including electrode properties like impedance, corrosion resistance and Q_{inj} [63,64]. Doses that cause clinically significant tissue damage will be location and tissue dependent and should be evaluated on a case-by-case basis. The data presented herein also have practical implications in vivo, as the DC observed in-vitro is typically much smaller than in-vivo [59]. With several reports indicating that chronic stimulator leakage currents in excess of 100 nA are associated with higher levels of tissue damage; by properly monitoring and mitigating them we can appreciably improve preclinical and clinical device safety while also maximizing longevity [22,59,65].

4.6 Limitations & Future Directions

When interpreting the results of this study, several limitations should be considered. The platinum electrodes used in this study were hand-fabricated using materials from various manufacturers. Although care was taken to standardize the surface condition, variability in surface morphology and potential contamination remain possible sources of error. Additionally, the absence of pre-trial control over V_{IPP} and electrolyte reactant concentrations may have contributed to some of the variability in Q_{inj} across trials. In some voltage transient trials, it was observed that V_{IPP} did not completely return to pre-trial levels, particularly when large changes in V_{IPP} occurred; likely reflecting persistent changes in the local electrode environment. Future work could improve reproducibility by incorporating potentiostatic control of V_{IPP} prior to stimulation.

This study used the platinum electrolysis window as a reference framework for assessing Q_{inj} , consistent with many previous reports. However, it is well established that there are corrosion/noxious reactions at platinum electrodes within the water window (i.e. formation of reactive oxygen species, platinum oxidation) [40,48,62,64]. Since electrochemical reactivity and failure mechanisms are material-specific, future studies should assess whether the electrolysis window remains a valid framework for evaluating Q_{inj} across emerging neuromodulation materials.

This study also tested a very specific set of parameters (i.e. pulse widths, amplitudes, frequencies) on the observed DC offset. Future experiments should explore the effects of waveform parameterization on DC offset magnitude and evaluate how sustained DC offsets affect electrode performance across different neuromodulation materials. Understanding these material-specific failure modes is critical for assessing long term safety and reliability of neuromodulation devices. In parallel, future studies should also characterize the effectiveness of DC mitigation strategies across these materials to determine whether existing approaches provide adequate protection or require adaptation for specific electrode compositions and geometries. Lastly, while we demonstrated the implications of electrode size on probe loading effects, the smallest electrodes used in this study were considerably larger than what is typically considered a microelectrode ($< 2 \times 10^{-5} \text{ cm}^2$). With these electrodes we

found that $10G\Omega - 10T\Omega$ of input impedance was sufficient to properly resolve the DC electrode potential, but additional research should be performed to understand the measurement instrumentation requirements TFMEAs.

Table 2. Key recommendations for reducing instrumentation confounds across neuromodulation studies.

Potential Issue	When is it a Concern?	Remediation/Reporting Recommendations	Instrumentation Concerns
Neuroscience Experiments			
DC offset accumulation at an electrode may result in a variety of undesirable side effects during stimulation. These include, but are not limited to: • water electrolysis • pH changes during stimulation • electrode corrosion • DC current mediated effects on tissue (i.e. damage, altered excitability)	Use of microelectrodes or small electrodes, of sizes commonly used in rodent studies.	Stimulator/electrode combinations used in preclinical experiments should be tested to confirm they do not cause excessive accumulation of DC offsets. Methods used for this validation should be reported.	Oscilloscopes commonly used to check stimulation waveforms mask DC offsets accumulation at electrodes. A high input impedance measurement system, that is appropriate for the DC impedance of the electrode is needed to appropriately measure accumulation of DC offsets.
	Materials with low availability of faradaic reactions (e.g. gold, titanium nitride)	Methods used to mitigate DC leakage currents at a stimulator, and their effectiveness, should be reported.	
		If accumulation of a DC offset occurs during experiments, it should be reported.	
		Stimulation Material/Electrode Development	
Instrumentation used for measuring charge injection limits (Q_{inj}) can lead to either over or underestimating the real charge injection limits of an electrode.	Care should be taken to ensure instrumentation is sufficient for small electrodes and non-faradaic materials.	The specifics of the test instrumentation used should be reported with sufficient detail to be replicated.	Measurement system input impedance must be much larger than the DC impedance of the electrode. (>100X)
			Stimulator output impedance must be much larger than the DC impedance of the electrode.
		The DC electrode impedance recorded around the open circuit potential should be reported so that reviewers can assess whether appropriate instrumentation was used.	Stimulator leakage currents should be mitigated such that they do not impact DC accumulation at the electrode.
			Steps should be taken to ensure parasitic impedance from cabling or circuitry, used for DC mitigation, is not lower than the DC electrode impedance. (see Supplementary Methods)

Neurostimulator Development

Neurostimulators do not commonly report the leakage current assessed across the full range of current outputs, the output impedance of the device, or methods used to mitigate DC leakage currents. As such, it is often difficult to assess if a stimulator is appropriate for the electrodes being used.	Care should be taken to ensure instrumentation is sufficient for small electrodes and non-faradaic materials.	Leakage current assessed across the full range of current outputs should be reported.	
		The output impedance of the stimulator assessed across an appropriate frequency range should be reported.	
		Methods used to mitigate DC leakage currents should be reported.	

Neuromodulation System (Stimulator + Electrode) Development

When developing a neuromodulation system that includes both a stimulator and electrode, the implicit Q_{inj} of the material or electrode used for stimulation may not be achieved due to limitations of the stimulator.	Care should be taken when using stimulators that were not designed for or previously validated for the electrode size and materials being used. For instance, stimulating microelectrodes with an IPG originally designed for clinical macroelectrodes.	Q_{inj} should be assessed using the stimulator used for the neuromodulation system, so that the impact of the stimulator can be determined.	Measurement system input impedance must be much larger than the DC impedance of the electrode.
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5. Conclusion

This study examined the role of measurement system impedance and DC mitigation strategies in accurately determining Q_{inj} using voltage transients. Our findings demonstrate that insufficient input impedance leads to significant underestimation of V_{IPP} shifts and, consequently, overestimation of Q_{inj} . These errors can obscure the onset of irreversible faradaic processes, misrepresenting electrode safety and compromising the validity of electrochemical testing.

Unmitigated stimulator leakage currents pose serious risks to long-term device performance and tissue safety. Sustained V_{IPP} drift has been shown to accelerate corrosion, alter local pH and promote the accumulation of toxic reaction byproducts. These risks are especially pronounced in microelectrodes, which have reduced tolerance to electrochemical imbalance before critical failures. Unmitigated/uncharacterized DC can also change dose response curves for specific electrophysiological outcomes, potentially causing poor reproducibility of key physiological findings across studies [24].

As such, we emphasize the need for standardized reporting of instrumentation parameters including input impedance, stimulator leakage currents and implemented DC mitigation strategies when reporting Q_{inj} . Electrode properties like R_{ct} and R_s should also be disclosed to establish the influence of measurement configuration on observed V_{IPP} values. **The accuracy and reproducibility of Q_{inj} measurements depend not only on the electrode material but on the entire electrochemical system and methodology used to interrogate it.**

Future work should focus on developing scalable and well-validated DC mitigation strategies suitable for characterizing multichannel implantable systems and on improving community-wide reporting standards to ensure that safety metrics can be meaningfully compared across labs, devices and clinical applications.

Conflicts of Interest

KAL is a co-founder and equity holder for Neuronoff, Inc. KAL is also a co-founder and equity holder of NeuraWorx. KAL is a scientific board member and has stock interests in NeuroOne Medical Inc. KAL is also a paid member of the scientific advisory board of Abbott, LivaNova and Presidio Medical, and a paid consultant for the Alfred Mann Foundation, ONWARD and Restora Medical. JKT is a paid consultant for Presidio Medical, Inc. LR is principal investigator on industry sponsored research contracts from Blackrock Neurotech.

Data Availability

All data that support the findings of this study are included within the article (and any supplementary files). Any additional data are available from the corresponding author upon reasonable request.

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